

Cross-Panel Shared-Nuisance Meta-Certification for a Finite Twin-Prime Response Model

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2 September 2026

Abstract

We perform a finite, protocol-locked meta-audit of two independent panels in a twin-prime response model. The first panel is the TPC-341 fresh holdout experiment and the second is its disjoint TPC-342 reproduction. A row-block nuisance projection, which permits one coefficient vector per source row, retains only 0.2325429101 of the pooled in-sample energy. In contrast, a single shared nuisance coefficient vector retains 0.3198013104 under raw energy weighting and 0.3549335801 after equal-row normalization, both above the inherited 0.30 guard. Nine shared cross-panel leave-one-control-out tests retain 0.6408306196–0.9090948298. These results certify a scoped finite obstruction to one particular shared nuisance law, while preserving the distinction between exact finite linear algebra and the still-open source-uniform arithmetic estimates required by the twin-prime route.

1 Question and scope

The preceding TPC-341 experiment projected a nine-control twin-prime mean onto the span of three nuisance means (non-twin prime shift, prime-power shift, and zero support). Its TPC-342 successor repeated the same protocol on a new disjoint source panel. Both experiments showed a strong in-sample fit and a much weaker omitted-control fit. The next minimal question is whether the two panels nevertheless share one nuisance coefficient law.

This paper answers only that finite model-comparison question. The words “nuisance” and “twin” label declared masks and vectors; they do not assert that the nuisance directions are arithmetic noise. No asymptotic estimate, probability statement, or twin-prime theorem is claimed.

2 Frozen finite protocol

The all-plus deleted-diagonal operator has $Q = 54$, kernel exponent one, and height $H = 66$. Each source row has length 512 and is formed from a window of scale 1024. The nine coordinate controls are the identity, reversal, and the seven stated affine permutations from the TPC-338/TPC-340 orbit. The two panels are

$$\begin{aligned}\mathcal{P}_{341} &: [48097, 48608], [48609, 49120], [49217, 49728], \\ \mathcal{P}_{342} &: [40097, 40608], [40609, 41120], [41121, 41632].\end{aligned}$$

All shifted source arguments are below the locked cutoff 50000. The six rows produce $6 \cdot 9 \cdot 4 = 216$ raw records, of which 171 are nonempty, and $6 \cdot 9 = 54$ omitted-control records.

For row r , let $y_r \in \mathbb{R}^{512}$ denote the twin-prime mean output and let $N_r = (n_{r,1}, n_{r,2}, n_{r,3})$ denote the nuisance means. We compare:

1. the row-block model, whose nuisance matrix is the orthogonal direct sum of the six N_r ;
2. the shared model, whose three columns are the vertical concatenations $(n_{1,j}, \dots, n_{6,j})$ for $j = 1, 2, 3$.

The first model has row-specific coefficients. The second forces one coefficient vector across the whole panel. To test sensitivity to scale, the shared model is also evaluated after dividing each row's target and all of its nuisance columns by $\|y_r\|_2$.

3 Finite identities

Let N be any finite real matrix and let P_N be the Euclidean orthogonal projector onto its column space.

Proposition 1 (stacked Pythagorean identity). *For every finite vector Y ,*

$$\|Y\|_2^2 = \|P_N Y\|_2^2 + \|(I - P_N)Y\|_2^2. \quad (1)$$

Proof. The two terms on the right are in the column space of N and its orthogonal complement, respectively, and their sum is Y . $\square$

For the row-block matrix, the column supports of different rows are disjoint. Thus its projector is the direct sum of the row projectors, and both projected and residual energies add exactly. This is why the row-block pooled retention is an energy-weighted finite average of the six row retentions. The shared matrix has only three global columns and is a different declared span. The equal-row transformation changes only the meta weight, not the within-row projection geometry.

As a small exact anchor, take

$$Y = (1, 1, 1, 1)^T, \quad n = (1, 0, 1, 0)^T.$$

The projection of Y onto $\text{span}\{n\}$ has energy 2, as does the residual; hence $\|Y\|_2^2 = 2 + 2$ and the residual retention is exactly $1/2$. This anchor is independent of the numerical source calculation.

4 Independent computational audit

The producer uses the hash-locked TPC-340 source/operator implementation. A separate reverse-shell checker uses the hash-locked TPC-340 independent engine and reimplements source classification, all nine permutations, the two leave-one-control-out loops, and both stacking constructions. The TPC-341 and TPC-342 producer and certificate hashes are checked before either panel is read. A seven-mutation stress suite rejects changes to row geometry, meta guards, identity metadata, and the arithmetic firewall.

The exact finite census is summarized in Table 1.

The row-block model passes the inherited in-sample guard $\rho < 0.30$. Both shared variants fail that guard. The failure is not caused by a single outlying row: the panel-specific shared retentions are

The shared raw and equal-row singular-value condition diagnostics are about 192.25 and 320.17, respectively. They are reported to make the finite coordinate choice visible; they are not a conditioning theorem.

Table 1: Frozen census and pooled finite readout.

quantity	value
panels / rows	2/6
raw / nonempty records	216/171
in-sample projections	6
leave-one-control-out projections	54
row-block raw retention	0.2325429101
row-block equal-row retention	0.2502856954
shared raw retention	0.3198013104
shared equal-row retention	0.3549335801
weighted individual holdout retention	0.6669022118

Table 2: Panel-specific in-sample meta retentions.

panel	row-block	shared raw	shared equal-row
TPC-341	0.2106754608	0.2537762890	0.2748728968
TPC-342	0.2735950585	0.3758972339	0.3624284431
combined	0.2325429101	0.3198013104	0.3549335801

For each omitted control, we stack the six held-out twin outputs and the corresponding eight-control nuisance means. The shared raw residual retention lies in $[0.6408306196, 0.9090948298]$. The individual, row-local held-out range across all 54 records is

$$[0.4435267486, 0.9429165296].$$

The holdout guard is therefore satisfied in the declared direction, while it does not rescue the failed shared in-sample law.

5 Interpretation and claim firewall

The strongest positive result is a reproducible finite distinction: row-local nuisance projection remains strong after pooling two panels, but the same three coordinates do not support one shared coefficient vector under either of two declared weightings. The strongest obstruction is thus a cross-panel coefficient-stability failure, not a failure of the orthogonal projection identity.

The status of the claims is:

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TPC343_STACKED_IDENTITY = PROVED_EXACT_FINITE_DECLARED_MODEL
TPC343_ROW_BLOCK_META = NUMERICALLY_CERTIFIED_FINITE_6_ROW_POOLED_PROJECTION
TPC343_SHARED_COEFFICIENT_STABILITY = REFUTED_SCOPED
TPC343_HOLDOUT_META = NUMERICALLY_CERTIFIED_FINITE_54_RECORDS
TPC343_ARITHMETIC_ADVANCE = NO
TPC343_FIXED_POWER_CREDIT = 0
TPC343_SOURCE_UNIFORM_L2 = OPEN
TPC343_FULL_GATE_B = OPEN
TPC343_TWIN_PRIME_RESULT = NONE

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The scoped refutation concerns only this shared basis and these two panels. It does not refute an alternative nuisance basis, and it cannot be promoted to a source-uniform arithmetic L^2 estimate or to an official Route-A/Route-B pass. The session-named evaluator files are absent in this checkout, so the repository records only a local fail-closed Bridge-B audit.

6 Next finite question

The natural next experiment is geometric rather than arithmetic: compare the principal angles of the row nuisance spans and test a predeclared alternative basis (or cross-fit basis) on a fresh panel. Any such result must retain the same finite/asymptotic firewall and earns zero fixed-power credit until an independent growing arithmetic estimate is supplied.